



Lab 02 - Class definition - Input & Output

A. Instructions

Requirements

1. Create a project named **FractionInput**
2. Define a **Fraction** class with two attributes:
 - Numerator: an integer
 - Denominator: an integer
3. Create a class diagram for this Fraction class (submit back an image inside your project)
4. Write a main function
 - a. Get a fraction from the keyboard
 - b. Print out the newly entered fraction

Sample code

Fraction.h

```
#pragma once class Fraction { private: int _num; int _den; public: int getNumerator(); int getDenominator(); void setNumerator(int value); void setDenominator(int value); public: Fraction(); public: void input(); void output(); };
```

Fraction.cpp

```
#include "Fraction.h" #include <iostream> using namespace std; int
Fraction::getNumerator() { return _num; } int
Fraction::getDenominator() { return _den; } void
Fraction::setNumerator(int value) { _num = value; } void
Fraction::setDenominator(int value) { _den = value; }
Fraction::Fraction() { _num = 0; _den = 1; } // Week02: Use input()
function // Week03: Use FractionKeyboardProvider.next() void
Fraction::input() { cout << "Nhap tu:"; cin >> _num; cout << "Nhap
mau:"; cin >> _den; } // Week02: use "output" function and inside
this function access directly to the console using cout // Week03:
use toString() // Week04: use Converter void Fraction::output() {
cout << _num << "/" << _den; }
```

main.cpp

```
#include <iostream> using namespace std; #include "Fraction.h" int
main() { cout << "Nhap xuat so nguyen" << endl << endl; cout <<
"Nhap vao mot phan so." << endl; Fraction a; a.input(); cout <<
"Phan so vua nhap la:"; a.output(); }
```

Result

Nhap xuat so nguyen Nhap vao mot phan so. Nhap tu:4 Nhap mau:5 Phan
so vua nhap la:4/5

B. Exercises

Create a solution named **Week02**

1. Point in 2D space

1. Create a project named **PointInput**
2. Define a **Point** class with two attributes:
 - X: a float
 - Y: a float

3. Create a class diagram for this class (submit back an image inside your project)
4. Write a main function
 - a. Get a point from the keyboard
 - b. Print out the newly entered Point

2. Line

1. Create a project named **LineInput**
2. Define a Line class with four attributes:
 - X1: a float
 - Y1: a float
 - X2: a float
 - Y2: a float
3. Create a class diagram for this class (submit back an image inside your project)
4. Write a main function
 - a. Get a line from the keyboard
 - b. Print out the newly entered line

3. Date

1. Create a project named **DateInput**
2. Define a Date class with three attributes:
 - Day: an integer
 - Month: an integer
 - Year: an integer
3. Create a class diagram for this class (submit back an image inside your project)
4. Write a main function
 - a. Get a date from the keyboard
 - b. Print out the newly entered date

4. Student

1. Create a project named **StudentInput**
2. Define a Student class with five attributes:
 - ID: a string to represent the student's id
 - Fullname: a string
 - Address: a string
 - Email: a string
 - DOB: a Date to represent the student's date of birth
3. Create a class diagram for this class (submit back an image inside your project)
4. Write a main function
 - a. Get a student's information from the keyboard
 - b. Print out the newly entered student

5. Product

1. Create a project named **ProductInput**
2. Define a Product class with four attributes:
 - ID: a string
 - Name: a string
 - Price: a float
 - ExpiryDate: a Date
3. Create a class diagram for this class (submit back an image inside your project)
4. Write a main function
 - a. Get a product's information from the keyboard
 - b. Print out the newly entered product

C. Submission instructions

1. Delete the hidden folder **.vs**
2. Compress into the format {StudentID}.zip/.rar and submit back this file.

- For example if your student id is 1812445, your filename should be 1812445.zip or 1812445.rar

D. Frequently asked questions

No question yet.